

LAB REPORT

DEPARTMENT : SCSET

CSET 101 : COMPUTATIONAL THINKING AND PROGRAMMING

BUDGET BUDDY

INTRODUCTION :

A Budget Management System is a tool designed to help users track income and expenses, allocate budgets, and analyze financial habits. It simplifies financial planning by providing insights through reports and visualizations, promoting financial discipline, transparency, and efficiency. This system ensures secure, accurate, and user-friendly management of personal or organizational finances.

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TEAM MEMBERS WITH THEIR WORK DISTRIBUTION :

- Samridhi Agrawal : Created sign-up form and displayed data.
 - Sanjana Kumari : Managed the data that would be entered in the file.
 - Riya Kabra : Created line graph for better visualization.
 - Sahaj Khanna : Created bar graph for choosen week for better visualization.
 - Anubhuti Singh : Created a pie chart for monthly expense analysis.
 - Sheshank Swamy : Modified the program using toolkit interface.
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PROBLEM STATEMENT :

Managing finances effectively is a challenge faced by individuals and organizations due to the lack of efficient tools for tracking income, expenses, and budgeting. Many rely on manual methods or generic applications that fail to provide tailored insights, leading to poor financial planning, overspending, and difficulty in achieving financial goals. This project aims to address these issues by developing a comprehensive Budget Management System that simplifies financial tracking, enhances budget allocation, and empowers users to make informed financial decisions.

SOLUTION APPROACH :

Development of a Comprehensive Budget Management System :

- Core Implement features like real-time income and expense tracking, customizable budget categories.
 - Provide visualizations (line and bar graphs, pie charts) to help users understand spending patterns and identify areas to save.
 - Develop the system with robust data encryption to ensure user confidentiality and prevent unauthorized access.
 - Focus on intuitive design with clear navigation and simplified processes for recording and analyzing financial data.
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TECHNOLOGIES USED :

- **Programming language:** python
 - **Tkinter:** For GUI development.
 - **Matplotlib:** For plotting graphs and visualizing data.
 - **OS:** For file handling.
 - **Datetime:** To manage date and time operations.
 - **AST (Abstract Syntax Trees):** To safely parse strings into Python objects.
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FEATURES :

- Account creation with unique user id
 - Secured id with password
 - Expense history
 - Track daily, weekly, monthly expenses
 - Trace daily expenditure through line graph
 - Monitor weekly expense through bar graph
 - Record monthly expenditure through pie chart
 - View expense of a particular day from the past
 - The generated graph gets stored in the device
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CODE STRUCTURE :

1. SYSTEM WORKFLOW :

- Login/Sign Up page
- In case of incorrect password, a message box displays invalid password
- Add daily expense
- Day/Week end
- View expense history
- View total expense of day/week/month
- Daily line graph
- Weekly bar graph
- Monthly pie chart
- Logout functionality

2. FILE MANAGEMENT :

- **User Credentials:** Stored in passdict.txt. Each user has a unique file (userid.txt) to store their expense data.
- **Expense Data Format:** Stored in a nested structure, representing weeks and days as follows:

```
(['week1'], [['Monday'], {'item1': [amounts], 'item2': [amounts]}], ['Tuesday'], {'item1': [amounts]}])
```

3. GRAPHS :

- Matplotlib generates:
 - Pie charts for monthly summaries.
 - Bar graph for weekly summaries.
 - Line graphs for daily summaries.
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USAGE INSTRUCTION :

1. RUNNING THE PROGRAM :

- Save the code in a Python file (budget_buddy.py).
- Run the file using python budget_buddy.py.

2. FEATURES IN DETAILS :

Signup/Login

- Enter a User ID and Password.
- If new credentials are entered, an account is created.
- If existing credentials are provided, the program validates them.

Add Expenses

- Specify an item name and the amount.
- The application updates the current day's expenses in the file.

View Expenses

- Displays all expenses stored in the user's file, categorized by week and day.

End Day/Week

- Creates new entries in the file for a new day or week.

Display Total

- Summarizes expenses:
 - Daily Total: For a specific day of a given week.
 - Weekly Total: Aggregates across all days in a given week.
 - Monthly Total: Aggregates across all weeks.

Graph Analysis

- Visualizes spending:

Day Analysis: Pie chart of categories.

Week Analysis: Bar graph of total spending per category.

Month Analysis: Pie chart of monthly spending.

CONCLUSION :

This budget management system enables efficient expense tracking and financial analysis, providing users with visual insights through graphs to support data-driven budgeting decisions and optimize spending patterns.

Strengths

- User Authentication: Robust account creation and login functionality.
- Data Persistence: Expense data is securely stored and retrievable.
- Dynamic Graphs: Provides intuitive visualizations for better analysis.
- User-Friendly Interface: Simple and effective use of Tkinter widgets.

Limitations

1.File-Based Data Storage:

- A database would be more efficient for scaling.

2.Data Integrity:

- Manual editing of data files may cause errors.

Improvements

1.Database Integration:

- Replace file-based storage with a database (e.g., SQLite or PostgreSQL) for better performance and scalability.

2.Enhanced Input Validation:

- Add robust validation for user inputs.

3.UI Improvements:

- Enhance design with frameworks like PyQt.

4.Testing:

- Add unit tests to validate core functionalities.
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